

flodym (Flexibe Open Dynamic Material Systems Model) library overview

The flodym (Flexibe Open Dynamic Material Systems Model) library provides key functionality for building material flow analysis models, including

- the class MFASystem acting as a template (parent class) for users to create their own material flow models
- the class FlodymArray handling mathematical operations between multi-dimensional arrays
- different classes representing stocks accumulation, in- and outflows based on age cohort tracking and lifetime distributions. Those can be integrated in the MFASystem.
- different options for data input and export, as well as visualization

flodym (flexible ODYM) was developed by Jakob Dürrwächter and colleagues at PIK Potsdam. It is an adaptation of the ODYM MFA software library, by Stefan Pauliuk, Uni Freiburg, Germany. The code is already being used in a number of research projects.

Here's the code: <https://github.com/pik-piam/flodym>

And the documentation: <https://flodym.readthedocs.io/en/latest/>

Two of the creators' own applications:

<https://github.com/pik-piam/simson>

<https://github.com/pik-piam/remind-mfa>

For the 2026 Bath summer school on dynamic MFA modelling in Python, we focus on vehicle fleet modelling of the stock-flow-service nexus using the new flodym package. From transport demand and up to GHG and material footprint, both inflow-driven and stock-driven fleet models are used.

The exact setup of the sessions will be determined on the spot, depending on the dMFA understanding and coding experience of the colleagues. As a rough orientation, the following software exercises in notebook form are offered:

- **flodym_example1:** Basic MFA system to get started
- **flodym_Bath_dMFA_Global_Vehicle_Fleet_use_phase_1:** Inflow-driven dynamic stock model for the vehicle fleet. Two versions of the notebook are available: with and without partly self-completion. In the second part, we'll add a new technology dimension to separate the fleet into gasoline and battery electric vehicles.
- **flodym_Bath_dMFA_Global_Vehicle_Fleet_use_phase_2:**, build on `_use_phase_1`, but with stock-driven model and time horizon till 2060. Two versions of the notebook are available: with and without partly self-completion.
- **flodym_Bath_dMFA_Global_Vehicle_Fleet_material_cycle:** build on `_use_phase_2`, add material cycle flows via material composition and recycling parameters. Two versions of the notebook are available: with and without partly self-completion.
- **flodym_Bath_dMFA_Global_Vehicle_Fleet_material_cycle_energy:** build on `_material_cycle`, add energy and scope 1-2-3 GHG via new parameters and model equations. Two versions of the notebook are available: with and without partly self-completion.